A-A-54014
6 November 1989

COMMERCIAL ITEM DESCRIPTION

GINGIVAL RETRACTION CORD
(Impregnated with Racemic Epinephrine Hydrochloride)

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Military Specification MIL-G-36509C, which is cancelled.

This commercial item description covers impregnated gingival retraction cords suitable for use in gingival retraction in dental procedures.

Salient characteristics:

The gingival retraction cords shall be of the following types as specified below:

Type I. - 4 ply cord

Type II. - 2 ply cord

Material.

Type I. Type I cord shall be 4-ply, spun from bleached, virgin, long staple cotton. The finished cord shall be formed by twisting four lengths of single ply yarn 84 ± 3 turns per foot in a direction opposite to the twist of each single strand. The finished cord shall have a soft, expandable texture without a starch or polish finish.

Type II. Type II cord shall be 2-ply, spun from bleached, virgin, long staple cotton. The finished cord shall be formed by twisting two lengths of single ply yarn 24 ± 1 turns per foot in a direction opposite to the twist of each single strand. The finished cord shall have a soft, expandable texture without a starch or polish finish.

AMSC N/A FSC 6520

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Racemic epinephrine hydrochloride. The racemic epinephrine hydrochloride shall be of high pharmaceutical grade suitable for the purpose intended.

Description. The type I and type II cotton cords shall be impregnated with a buffered racemic epinephrine hydrochloride solution using a vacuum or roller press process, and shall be dehydrated to a dry stable state. Type I and type II retraction cords shall be white in color and shall both have a minimum length of 72 inches. Ten yards of type I untreated cotton cord shall weigh a minimum of 2.00 grams (2,268 yards per pound). Ten yards of type II untreated cotton cord shall weigh a minimum of 2.50 grams (1,810 yards per pound).

Physical properties.

Breaking strength. Type I retraction cord shall have a minimum breaking strength of 6.0 pounds, when tested in accordance with ASTM Standard D2256. Type II retraction cord shall have a minimum breaking strength of 7.0 pounds, when tested in accordance with ASTM D2256.

Epinephrine hydrochloride content. Type I and Type II retraction cord shall contain 0.45 to 0.55 milligram (mg) of racemic epinephrine hydrochloride per inch of cord, when tested as specified below.

Immediate container. The immediate container for the retraction cord shall be a plastic dispenser having a suitable closure to prevent contamination or deterioration of the cord. The immediate container shall not interact physically or chemically with the retraction cords.

Instructions for use. Adequate and accurate instructions for the use of the retraction cords shall accompany each unit package.

Shelf life. The shelf life of the retraction cord shall be not less than 36 months from date of manufacture.

Delivery schedule. Not more than 3 months shall have elapsed from the date of manufacture of the retraction cords to the date of delivery to the Government.

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Workmanship. Gingival retraction cords shall be free from defects which detract from their appearance or impair their serviceability. The immediate containers shall be free from all dirt, foreign matter, and other contaminants, both before and after filling.

Unit. Package (PG). One package containing the bottle of 72 inches of cord, and instructions for use, as specified, constitutes one unit.

Unit package. The bottle of retraction cord and instructions for use shall be packaged in a suitable box. The bottle and box shall be marked with the lot or control number and the expiration date.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the same sample shall be used for the determination of two or more inspection characteristics. The unit of product for inspection purposes shall be one package of retraction cord. Unless otherwise specified, sampling for examination shall be conducted at inspection level II, and sampling for tests (when applicable) shall be conducted at inspection level S-1.

Examination. The retraction cords shall be examined for defects including, but not limited to, those listed below.

Defects

Type of cord not as specified.
Color of cord not white as specified.
Texture of cord not as specified.
Length of cord not as specified.
Cord has other defects which detract from its appearance or impair its serviceability.
Immediate container not proper type.
Lot number and expiration date not marked on immediate container.
Immediate containers not clean.
Instructions for use missing, incomplete or illegible.

Tests. Tests shall be conducted to determine compliance with specification requirements.

Assay for racemic epinephrine hydrochloride.

Test reagents. Dissolve 1.5 grams of ferrous sulfate heptahydrate in 200 milliliters (ml) of distilled water to which has been added 1 ml of 1 N hydrochloric acid and 1.0 gram of sodium bisulfite. The amount of iron in 0.1 ml of this solution is sufficient for the determination of quantities of epinephrine up to 0.5 mg. For larger amounts of epinephrine, it is necessary to employ a proportionately higher concentration of ferrous sulfate. Prepare the ferrous sulfate-citrate solution by adding 0.5 gram of sodium citrate ($Na_3C_6H_5O_7.2H_2O$) to 10 ml of the first solution. This solution should be prepared fresh each day. To prepare the buffer reagent, add 42 grams of sodium bicarbonate and 50 grams of potassium bicarbonate to 180 ml of distilled water. Not all of the solids will be dissolved at this stage. To another 180 ml of water, add 37.5 grams of aminoacetic acid and 17 ml of strong ammonia solution (28 percent NH_3). Mix the two solutions and add sufficient water to adjust the volume to 500 ml. The remaining solids will be dissolved at usual room temperatures after the final addition of water.

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Assay test. Suspend a 5-inch length of retraction cord in 50 ml of distilled water and allow to soak at room temperature for 15 minutes. Remove a 10 ml aliquot with a pipet, place in a test tube and add 0.1 ml of ferrous-citrate reagent, followed by 1.0 ml of the buffer reagent. Mix the resulting solution and allow to stand for at least 10 minutes to develop full color. The color developed is read in a spectrophotometer at a wave length of 530 millimicrons. The concentration of racemic epinephrine hydrochloride is read directly from a calibration curve. The calibration curve is prepared by plotting optical density against the concentration of racemic epinephrine hydrochloride. The density readings are obtained after the instrument has been set to read 0.0 when the comparison tube contains distilled water. Three determinations per lot shall be conducted and the results shall be averaged.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pounds units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

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Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Numbers are covered by this document:

National Stock Number	Item Identification	Specification Designation
6520-00-890-1864	GINGIVAL RETRACTION CORD, IMPREGNATED, DENTAL, 4 Ply	Type I
6520-00-890-1865	GINGIVAL RETRACTION CORD, IMPREGNATED, DENTAL, 2 Ply	Type II

Ordering data. Intermediate/exterior package quantities, labeling and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

PROJECT NO. 6520-2293
LOCATION: NCR DISK 2

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-54014	GINGIVAL RETRACTION CORD (Impregnated with Racemic Epinephrine (Hydrocchloride)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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